

Lin Ai

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Education

Ph.D.	Columbia University Computer Science, Natural Language Processing Advisor: Prof. Julia Hirschberg	Sep 2020 – Oct 2025
M.Sc.	Columbia University Computer Science with Machine Learning Track	Sep 2018 – Feb 2020
B.Math	University of Waterloo Computer Science, Statistics, Actuarial Science Graduated with Distinction.	Sep 2013 – Apr 2018

Research Interests

Human-Agent Systems: Context-aware productivity agents, workflow/workstream understanding, user and agent telemetry, agent evaluation.

Trustworthy NLP: LLM safety and alignment, information disorder, social engineering defense, responsible AI.

Professional Experience

Microsoft, <i>Senior Applied Scientist</i> (IDEAS Research led by Dr. Scott Counts)	Mountain View, CA Nov 2025 – Present
- Advancing research on productivity agents that use workflow, workstream, artifact, collaborator, and context signals beyond isolated task-level assistance. - Developed telemetry-driven workflow understanding methods that convert Microsoft 365 user actions into hierarchical natural-language representations, enabling analysis across tasks, sessions, work episodes, and long-running workstreams. - Delivered research prototypes, internal demos, technical writeups, and evaluation frameworks for context-aware agent experiences, including workflow-centered memory, workstream discovery, and telemetry-based diagnosis of Copilot-assisted workflows.	
Adobe Research, <i>Research Scientist Intern</i> (Data Intelligence Team led by Dr. Vishy Swaminathan)	San Jose, CA May 2025 – Aug 2025
- Developed SteER, an interactive deep research agent that adaptively pauses to solicit user guidance, enabling more personalized and controllable long-form research workflows. - Designed a cost-benefit pause policy and persona-aware branch selection method to balance user alignment, information gain, exploration diversity, and interaction cost. - Delivered research prototypes, internal demos, human evaluation studies, and invention disclosure materials demonstrating improved alignment and focus over autonomous deep research baselines.	
JP Morgan Chase, <i>AI/ML Data Associate Research Intern</i> (ML Team led by Dr. Lidia Mangu)	New York, NY Jun 2024 – Aug 2024
- Developed NovAScore, an automated document-level novelty metric that combines atomic content units with salience weighting to evaluate information distinctiveness and reduce redundancy. - Delivered a patent-pending framework for document curation, training data selection, and retrieval/evidence ranking.	
Meta, <i>Research Scientist Intern</i> (Multimodal Team led by Dr. Florian Metze)	New York, NY May 2022 – Sep 2022
- Built multimodal sentiment classification models for short-form videos, improving cross-modal representation learning and achieving competitive performance with state-of-the-art baselines. - Designed a multimodal BYOL self-supervised learning framework to improve cross-domain representation adaptation.	

Doctoral Research

Columbia University, <i>Ph.D. Researcher</i> Advisor: Prof. Julia Hirschberg Dissertation: <i>Towards Trustworthy AI: Detecting, Understanding, and Mitigating Information Disorder</i>	New York, NY Sep 2020 – Oct 2025
Developed a detection–understanding–mitigation framework for trustworthy AI systems addressing information disorder across social media, multimodal content, and LLM-mediated communication.	

- Built models and datasets for misinformation, propaganda, radicalization, malicious intent, and audience perception, integrating textual, visual, social-network, and human-centered signals.
- Designed mitigation systems for LLM-era risks, including social-engineering defense, factuality-controlled generation, distinctiveness-aware curation, and human-in-the-loop agent steering.

Selected Publications

[1] **An Interactive Paradigm for Deep Research**

L. Ai, V. S. Bursztyn, X. Chen, J. Hirschberg, S. Mitra

The 43rd International Conference on Machine Learning (*ICML'26*)

[2] **TeamFusion: Supporting Open-ended Teamwork with Multi-Agent Systems** [Paper](#)

J. Liu, V. S. Bursztyn, **L. Ai**, H. Wang, S. Choudhary, S. Mitra, Q. Wu

The 64th Annual Meeting of the Association for Computational Linguistics (*ACL'26*)

[3] **A Review of Incorporating Psychological Theories in LLMs** [Paper](#)

Z. Liu, Z. Gong, **L. Ai**, Z. Hui, R. Chen, C. W. Leach, M. R. Greene, J. Hirschberg

The 19th Conference of the European Chapter of the Association for Computational Linguistics (*EAACL'26*)

[4] **Detecting Mental Manipulation in Speech via Synthetic Multi-Speaker Dialogue** [Paper](#)

R. Chen, W. Liang, Z. Gong, **L. Ai**, J. Hirschberg

The 16th International Workshop on Spoken Dialogue System Technology (*IWSDS'26*)

[5] **Personalized Attacks of Social Engineering in Multi-turn Conversations – LLM Agents for Simulation and Detection** [Paper](#)

T. Kumarage, C. Johnson, J. Adams, **L. Ai**, M. Kirchner, A. Hoogs, J. Garland, J. Hirschberg, A. Basharat, H. Liu

AI Agents Workshop at the 2025 Conference on Language Modeling (*AIA@COLM'25*)

[6] **Comparison-Based Automatic Evaluation for Meeting Summarization** [Paper](#)

Z. Gong, **L. Ai**, H. Deshpande, A. Johnson, E. Phung, Z. Wu, A. Emami, J. Hirschberg

The 26th Annual Conference of the International Speech Communication Association (*Interspeech'25*)

[7] **NovAScore: A New Automated Metric for Evaluating Document Level Novelty** [Paper](#)

L. Ai, Z. Gong, H. Deshpande, A. Johnson, E. Phung, A. Emami, J. Hirschberg

The 31st International Conference on Computational Linguistics (*COLING'25*)

[8] **Propalnsight: Toward Deeper Understanding of Propaganda in Terms of Techniques, Appeals, and Intent** [Paper](#)

J. Liu*, **L. Ai***, Z. Liu, P. Karisani, Z. Hui, M. Fung, P. Nakov, J. Hirschberg, H. Ji

The 31st International Conference on Computational Linguistics (*COLING'25*)

[9] **Defending Against Social Engineering Attacks in the Age of LLMs** [Paper](#) [Code](#)

L. Ai*, T. Kumarage*, A. Bhattacharjee*, Z. Liu, Z. Hui, M. Davinroy, J. Cook, L. Cassani, K. Trapeznikov, M. Kirchner, A. Basharat, A. Hoogs, J. Garland, H. Liu, J. Hirschberg

The 2024 Conference on Empirical Methods in Natural Language Processing (*EMNLP'24*)

[10] **Enhancing Pre-Trained Generative Language Models with Question Attended Span Extraction on Machine Reading Comprehension** [Paper](#) [Code](#)

L. Ai, Z. Hui, Z. Liu, J. Hirschberg

The 2024 Conference on Empirical Methods in Natural Language Processing (*EMNLP'24*)

[11] **TweetIntent@Crisis: A Dataset Revealing Narratives of Both Sides in the Russia-Ukraine Crisis** [Paper](#)

L. Ai, S. Gupta, S. Oak, Z. Hui, Z. Liu, J. Hirschberg

The 18th International AAAI Conference on Web and Social Media (*ICWSM'24*)

[12] **Combating the COVID-19 Infodemic: Untrustworthy Tweet Classification using Heterogeneous Graph Transformer** [Paper](#) [Code](#)

L. Ai, Z. Liu, J. Hirschberg

The 17th International AAAI Conference on Web and Social Media (*ICWSM'23 Workshop*)

Technical Skills

Methods: LLM agents, human-agent interaction, telemetry modeling, multimodal learning, graph neural networks, evaluation, user studies.

Tools: Python, PyTorch, Hugging Face, LangChain/LangGraph, SQL, Azure, Git.